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The Effect of Parameters on the Output Forces of the Integration of Electroadhesion and Electrostatic Actuation Considering Breakdown Voltage

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In the previous research, the integration of electrostatic adhesion and actuation was proposed as a promising technique for climbing robots and proved by several prototypes. However, the effects of design parameters on the adhesive and driving forces, considering the breakdown voltage, are still unclear. Using FEM simulation, this paper investigates the relationship between the parameters and the output forces with considering the limitation posed by the discharge or dielectric breakdown. The results show that the thicknesses of cover layer and the base layer influence the maximum force of adhesion as well as actuation. The force ratio between adhesion and driving forces is approximately proportional to the thickness ratio specifically defined in this paper. Finally, we propose a strategy to select parameters such as electrode pitch and thickness of each layer for given requirements on the robot performance.

1. Introduction

With the similar principle, electroadhesion and electrostatic actuation are compatible to each other, so the integration of the two techniques becomes a good choice for climbing robots. One of the possible configurations of the integration includes two identical electrode films with three-phase electrodes inside, as shown in Fig. 1(a). It can output driving force and adhesive force at the same time, and reserve the advantages of the two techniques such as light-weight, flexible, and thin. Fig. 1(b) shows one of its possible application forms as a crawler climbing robot [1].

To employ the integration of electrostatic adhesion and actuation in a robot, we have to decide the size parameters of the electrode film, such as the thickness of each layer and the pitch of the electrodes, for desired output forces. However, it is a challenge to estimate the influence of different parameters on the maximum forces, because the parameters not only directly affect the force strengths, but also limit the maximum voltage in the aspect of electrical breakdown.

By FEM simulation, this paper analyzes effects of the parameters on the maximum output forces (adhesive and driving forces) of the integration, considering the limitation of breakdown voltage, and then proposes a strategy to choose the parameters for the desired performance.

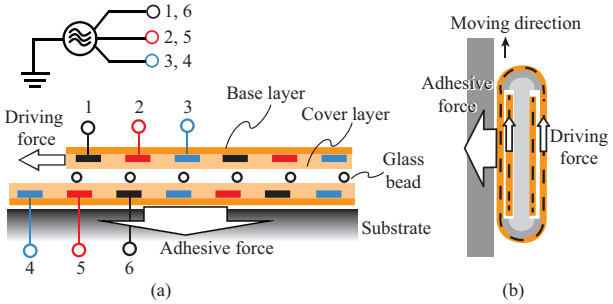


Fig. 1 (a) The configuration of the integration of electroadhesion and electrostatic actuation, and (b) its application to a crawler climbing robot

2. The effects of parameters on the output forces of the integration

1)The maximum adhesive force and driving force

This study analyzed the electrostatic forces and the maximum electric field through FEM analysis. The FEM model has a similar configuration shown in Fig. 1(a), and assumes that the electrode films are made by screen printing. In the model, the three parameters, the electrode pitch and the thicknesses of the cover layer and the base layer, are changed to examine their impact on the forces, while the other parameters are fixed considering the limitation of screen printing and the simplicity in analysis. The width of the electrodes is kept to the half of the electrode pitch, and the electrode thickness is 8 μm. Two air gaps exist: one gap of 50-μm thickness (including 20-μm glass beads) is between the two films; the other gap of 15-um thickness is just upon the substrate. Note that the thickness of cover layer refers to the distance from the cover layer surface to the electrode surface. The permittivity of the cover layer is set to 2.24, and the base layer 3.4.

We estimated the maximum forces under different parameters as followings. First, we calculated capacitance variation C_a between stator and slider electrodes by FEM, which was then converted into driving force using the following relation [2],

$$f_{dr} = \frac{3\pi}{2p} C_a V_0^2 \quad (1)$$

where p is the electrode pitch, C_a is the variation of the capacitance between one phase of stator and that of slider, and V_0 is the amplitude of the applied 3-phase voltage. Adhesive force, on the other hand, was directly calculated by FEM. The results are shown in Fig. 2 for the voltage amplitude of 1 V.

Next, we estimated the maximum voltage we can apply to the motor without having electric discharge based on the following assumptions on breakdown strength: the base layer is polyimide and has the breakdown strength of 384 MV/m; the electrical breakdown in the air gaps follows Paschen's curve, so the breakdown strength in the upper air gap is 15 MV/m, and in the lower air gap 27 MV/m. The resultant actuation voltage to cause breakdown in the films is higher than tens of kV, while the voltage to cause discharge in the air gaps is around several kV. Therefore we paid attention only to the air gap discharge. Fig. 2 shows the breakdown voltages calculated for the two air gaps respectively. Based on the forces calculated for 1V and the breakdown voltage, we can estimate the maximum adhesive and driving force.

Fig. 3(a) shows the maximum adhesive force for different cover layer and base layer thicknesses. When the cover layer thickness is thin compared to the base layer, it limits the maximum adhesive force because of the lower breakdown voltage in the air gap between films. If the cover layer becomes thicker, on the other hand, it has no effect on the adhesive force which is only limited by the base layer thickness. This figure plots the maximum adhesive force only for the electrode pitch of 300 μm, and the maximum force for the larger electrode pitch outcomes the close value due to the constant performance as shown in Fig. 2. This characteristic can facilitate parameter selection, which will be discussed in the next section.

Similar to the maximum adhesive force, the maximum driving force shown in Fig. 3(b) is limited by the base layer thickness when the base layer is comparably thin, and it only depends on cover layer thickness when the base layer is thick enough. However, what is different from adhesion is that the driving forces for different pitches are totally different due to the variance of force (for 1V) at different pitches shown in Fig.

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2(b).

2)The linear relationship between the force ratio and the thickness ratio

To facilitate the selection of parameters, this section discusses a relationship between the force ratio (adhesive force/ driving force) and the thickness ratio, which depends on the thicknesses of cover layer and the base layer. The paper uses ε_a, ε_c and ε_b to present the relative permittivity of the air, the cover layer and the base layer, and t_u, t_l, t_c and t_b for the thicknesses of the upper air gap, lower air gap, the cover layer and the base layer, respectively.

In the integration, adhesive force is approximately inversely proportional to $(\varepsilon_a t_b + \varepsilon_b t_l)^2$, and driving force roughly owns the linear relationship with the reciprocal of $(2\varepsilon_a t_c + \varepsilon_c t_u)$, so we define the thickness ratio as

$$r = \frac{2\varepsilon_a t_c + \varepsilon_c t_u}{(\varepsilon_a t_b + \varepsilon_b t_l)^2} \quad (5)$$

which contains all the thickness variables in the model.

Fig. 4(a) plots the force ratio with respect to the thickness ratio r with fitted lines. The force ratio grows proportionally to the thickness ratio, with determination coefficient R^2 being higher than 0.98. Note that this linear relationship is valid only for the pitches larger than 300 µm, and smaller pitch out of the range is subject to more errors.

3. Parameter selecting strategy

We can use the two subfigures in Fig. 4 to select the suitable parameters for the integration of electrostatic adhesion and actuation in a robot. As an example, this section chooses parameters for a crawler robot shown in Fig. 1(b) for the minimum payload of 2 N (including its weight) and adhesive area of 0.02 m^2.

First, we select the electrode pitch and the thickness ratio based on the desired force ratio and the lines in Fig. 4(a). The payload of a climbing robot is limited by the adhesive friction force $F_f = \mu f_{ad}$, and the total driving force $F_{dr} = k f_{dr}$, where f_{ad} is adhesion force, μ is the coefficient of friction, and k is the multiplying factor considering that the robot has two driving electrodes on both surfaces. Considering the tendency of the adhesive force and driving force, shown in Fig. 2, the optimal option is that the two forces equals each other; otherwise one of the force capability cannot be fully employed for the robot. Hence we have

$$\mu f_{ad} = k f_{dr} \quad (4)$$

In the specific situation, we set μ as 0.2, and k as 2 considering the robot has two identical sets of actuators. Therefore, the force ratio (f_{ad}/f_{dr}) should be 10, and according to Fig. 4(a), if the pitch is selected as 300 µm the thickness ratio should be around 0.022 μm^{-1}.

Then we continue to select the thicknesses of the layers according to Fig. 4(b). This figure shares the similar data with Fig. 3(a) but just replaces the simple variable of cover layer thickness into thickness ratio by Eq. 3 for parameter selection, such that it can be used for all the electrode pitches larger than 300 µm as explained above. Along the vertical line with the desired thickness ratio, we choose the parameter with adhesive force higher than 0.5 kN/m^2 due to the required payload. Here is one of the possible choices: electrode pitch is 300 µm, cover layer is 10-µm thick, and the base layer is 25-µm thick; many other parameters are also available by trading off according to Fig. 4. Although pitch smaller than 300 µm is unavailable for the figure, we can calculate it specifically based on the methods introduced before.

4. Conclusion

This paper analyzed the maximum force in the integration of electrostatic adhesion and actuation based on FEM considering the breakdown voltage. The maximum adhesive force is limited by the cover layer thickness when the cover layer is thin, but when the cover layer is thick enough it only depends on the base layer thickness. The maximum driving force has the similar relationship with the thicknesses of layers.

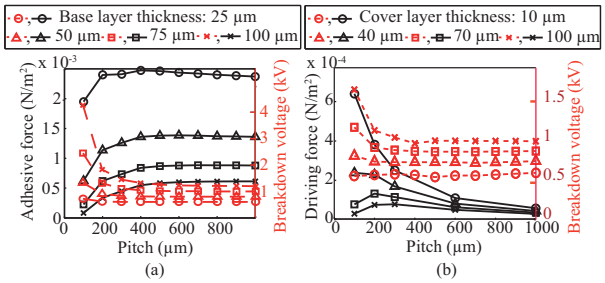


Fig. 2 (a) The adhesive force (under 1V) and the breakdown voltage in the air gap between the electrode film and the substrate; (b) the driving force (under 1V) and the breakdown voltage in the air gap between the electrode films

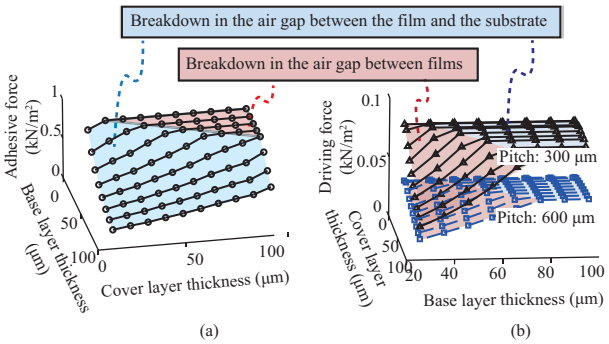


Fig. 3 (a) The maximum adhesive force with the effect of thicknesses of base layer and cover layer considering breakdown voltage (the pitch is 300 µm), and (b) the maximum driving force with the effect of thicknesses of base layer and cover layer considering breakdown voltage.

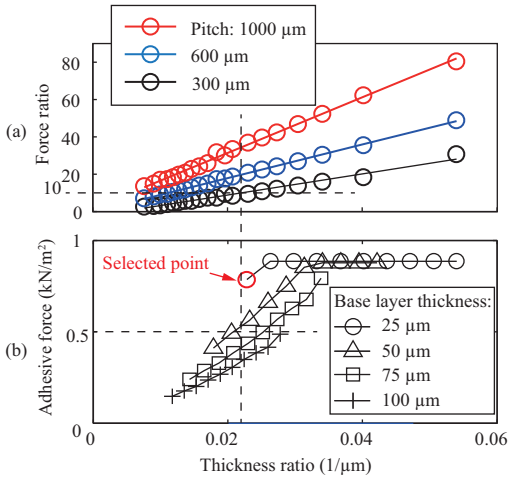


Fig. 4 The figures used for parameter selecting including (a) the force ratio with respect to the thickness ratio, and (b) the adhesive force under different thickness ratios

Besides, the force ratio is proved to be proportional to the thickness ratio of the layers. Based on the analyses, the paper demonstrated a parameter selection for the desired performance of application.

Reference

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